Statistical Mechanics

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13th September 2026

Entropy and Conservation of Information

Phase Space consists of positions and momenta. It is always even-dimensional. For N particles in a three-dimensional space, the phase space is $\mathbf{R}^{6\mathrm{N}}$.

Microstate is all the exact physical information about a system (all precise positions and momenta).

Macrostate is the easily observable data: internal energy E, volume V, and number of particles N. It is the collection of all microstates that share these same characteristics.

Conservation Law. The phase space divides into cycles, and states remain inside them.

Conservation of Information. Information is never lost; we always know where we came from. Every state has exactly one arrow out and one arrow in.

Liouville's Theorem. The volume of any region in the phase space will stay the same over time. If you start with a uniform distribution, it stays uniform.

Entropy is more fundamental than energy and temperature. It measures the number of hidden states that have a nonzero probability. It's dimensionless.

In [[Information Theory]], entropy is minus the sum of the probabilities times the log of the probabilities:

$$\mathrm{S} = -\sum_i \mathrm{P}(i) \log \mathrm{P}(i) = -\langle \log \mathrm{P}(i) \rangle.$$

Example: Think about n tossed coins. The total number of states is $\mathrm{N} = 2^n$. If we know nothing, the entropy is $\mathrm{S} = n \log_2 2 = n$ **bits**. A bit is the basic unit of entropy for a system with only two states.

In physics (thermodynamics), we add Boltzmann's constant k to give it physical units, assuming all Ω microstates are equally likely:

$$\mathrm{S} = k \log \Omega.$$

To measure Ω we integrate the volume over the measure $d^3x_1 d^3p_1 \cdots d^3x_\mathrm{N} d^3p_\mathrm{N}$. In order for these numbers to be dimensionless, we divide them by Planck's constant. Since all N particles are indistinguishable, we divide them by N!. Thus,

$$\Omega = \frac{1}{\mathrm{N}!} \frac{1}{h^{3\mathrm{N}}} \int_{\mathrm{Vol}} \prod_{i=1}^{\mathrm{N}} d^3x_i d^3p_i.$$

The zeroth law of thermodynamics. If system A is in thermal equilibrium with B, and B is in thermal equilibrium with C, then A is in thermal equilibrium with C.

The first law of thermodynamics. Energy conservation, nothing more. The time derivative of the total energy equals zero.

Temperature

Temperature is a derived notion that starts with energy and entropy. We define the **temperature** T of the macrostate as

$$\frac{1}{\mathrm{T}} = \frac{\partial \mathrm{S}}{\partial \mathrm{E}},$$

where E is the internal energy of the system. Temperature tells you how much the entropy changes when you add or remove energy.

Notice that if the system has a low temperature, a little increase in energy makes the entropy jump up by a lot.

Suppose that two systems A and B are put together, free to exchange energy. Since the total number of microstates in the system AB is $\Omega_A\Omega_B$, the total entropy S_{AB} is

$$S_{AB} = k\log(\Omega_A\Omega_B) = k\log(\Omega_A) + k\log(\Omega_B) = S_A + S_B.$$

The Second Law of Thermodynamics (also called **Law of Increase of Entropy**) states that an isolated system will naturally progress towards the macrostate with the largest number of microstates (highest entropy). Therefore, the entropy will increase until it hits its maximum.

Since the total energy is fixed, any energy gained by A is lost by B, that is $\Delta E_A = -\Delta E_B$. The total entropy stops changing (reaches its maximum) when its derivative with respect to E_A is zero:

$$0 = \frac{\partial S_A}{\partial E_A} + \frac{\partial S_B}{\partial E_A} = \frac{\partial S_A}{\partial E_A} - \frac{\partial S_B}{\partial E_B} = \frac{1}{T_A} - \frac{1}{T_B} \implies T_A = T_B.$$

In plain English, the system reaches its equilibrium when the temperatures are equal.

We also define the **pressure** P and the **chemical potential** μ as follows:

$$\frac{P}{T} = \frac{\partial S}{\partial V} \quad \text{and} \quad \frac{\mu}{T} = -\frac{\partial S}{\partial N}.$$

In equilibrium, we also have $P_A = P_B$ and $\mu_A = \mu_B$.

Maximising Entropy

Energy flows from higher to lower temperature. Temperature tells you which direction energy flows. This is the zeroth law. Let us prove this fact.

Let A and B be two systems that are allowed to exchange energy. Suppose wlog that $T_B > T_A$.

From the first law of thermodynamics,

$$dE_A + dE_B = 0.$$

Replacing the definition of temperature, i.e., $T = \frac{\partial E}{\partial S}$, one has

$$T_A dS_A + T_B dS_B = 0 \iff dS_B = -\frac{T_A}{T_B} dS_A.$$

The second law of thermodynamics says that

$$dS_A + dS_B \geq 0,$$

which yields, after replacing our previous identity and multiplying by T_B,

$$T_B dS_A - T_A dS_A > 0 \iff (T_B - T_A) dS_A \geq 0.$$

Since $T_B > T_A$, one has $T_B - T_A > 0$ and so $dS_A \geq 0$. Multiplying by T_A, we have

$$T_A dS_A = dE_A \geq 0,$$

as desired.

Mathematically, our goal is to find a function (a probability distribution) that gives the maximum value $-\sum P_i \log P_i$, such that $\sum P_i = 1$ and $\sum P_i E_i = E$. To solve this, we use Lagrange multipliers.

The Boltzmann Distribution

Recall that our goal is the following optimisation problem

$$\begin{aligned} \min_{\mathrm{P}_i} \quad & \sum_i \mathrm{P}_i \log \mathrm{P}_i \\ \text{s.t.} \quad & \sum_i \mathrm{P}_i = 1, \\ & \sum_i \mathrm{E}_i \mathrm{P}_i = \mathrm{E}. \end{aligned}$$

Using the Lagrange multipliers, we want to minimise

$$\mathrm{F}(\mathrm{P}_i) := \sum_i \mathrm{P}_i \log \mathrm{P}_i + \alpha \left[\sum_i \mathrm{P}_i - 1 \right] + \beta \left[\sum_i \mathrm{E}_i \mathrm{P}_i - \mathrm{E} \right].$$

Differentiating with respect to P_i and setting to zero yields

$$\frac{\mathrm{dF}}{\mathrm{dP}_i} = \log \mathrm{P}_i + 1 + \alpha + \beta \mathrm{E}_i = 0,$$

i.e.,

$$\log \mathrm{P}_i = -(1+\alpha) - \beta \mathrm{E}_i,$$

and

$$\mathrm{P}_i = e^{-(1+\alpha)} e^{-\beta \mathrm{E}_i} = \frac{1}{\mathrm{Z}} e^{-\beta \mathrm{E}_i},$$

where $\mathrm{Z} := e^{1+\alpha}$.

We'll see that β is the **inverse temperature**.

To normalise the P_i's, we need

$$\sum_i e^{-\beta \mathrm{E}_i} = \mathrm{Z}.$$

Notice that Z is a function of β, and it's called **partition function**.

To determine β, we use the average energy:

$$\sum_i \frac{1}{\mathrm{Z}} e^{-\beta \mathrm{E}_i} \mathrm{E}_i = \mathrm{E}.$$

Notice that differentiating the partition function with respect to β and dividing by $-\mathrm{Z}$ yields

$$-\frac{1}{\mathrm{Z}} \frac{\partial \mathrm{Z}}{\partial \beta} = \sum_i \frac{\mathrm{E}_i e^{-\beta \mathrm{E}_i}}{\mathrm{Z}} = \mathrm{E}.$$

That is,

$$\mathrm{E} = -\frac{\partial \log \mathrm{Z}}{\partial \beta}.$$

Using the previous identities, we can rewrite the entropy as follows:

$$\begin{aligned} \mathrm{S} &= -\sum_i \mathrm{P}_i \log \mathrm{P}_i \\ &= \sum_i \frac{1}{\mathrm{Z}} e^{-\beta \mathrm{E}_i} (\beta \mathrm{E}_i + \log \mathrm{Z}) \\ &= \beta \mathrm{E} + \log \mathrm{Z}. \end{aligned}$$

This logarithm of the partition function Z when multiplied by $-k_{\mathrm{B}}\mathrm{T}$ is called **Helmholtz free energy**.

Differentiating the expression for the entropy above,

$$\mathrm{dS} = \beta\,\mathrm{dE} + \mathrm{E}\mathrm{d}\beta + \frac{\partial \log \mathrm{Z}}{\partial \beta}\mathrm{d}\beta = \beta\,\mathrm{dE}.$$

Therefore,

$$\frac{1}{\mathrm{T}} = \frac{\mathrm{dS}}{\mathrm{dE}} = \beta.$$

This is why β is called the **inverse temperature**.

The Ideal Gas

An ideal gas consists of N non-interacting point molecules of mass m in a box of volume V (e.g. a rarefied gas). The energy of the system is purely kinetic, meaning $\mathrm{E} = \sum \frac{p^2}{2m}$.

The partition function in this case is

$$\begin{aligned}\mathrm{Z} &= \int e^{-\frac{\beta}{2m}\sum_{n=1}^{3\mathrm{N}} p_n^2}\,\mathrm{d}^{3\mathrm{N}}x\mathrm{d}^{3\mathrm{N}}p \\ &= \frac{\mathrm{V}^{\mathrm{N}}}{\mathrm{N}!}\left(\int e^{-\frac{\beta}{2m}p^2}\,\mathrm{d}p\right)^{3\mathrm{N}}.\end{aligned}$$

Changing $\frac{\beta}{2m}p^2 = q^2$, the integral inside equals

$$\int e^{-\frac{\beta}{2m}p^2}\,\mathrm{d}p = \sqrt{\frac{2m}{\beta}}\int e^{-q^2}\,\mathrm{d}q = \sqrt{\frac{2m\pi}{\beta}}.$$

Hence,

$$\mathrm{Z} = \frac{\mathrm{V}^{\mathrm{N}}}{\mathrm{N}!}\left(\frac{2m\pi}{\beta}\right)^{3\mathrm{N}/2}.$$

Using Stirling's formula,

$$\frac{\mathrm{V}^{\mathrm{N}}}{\mathrm{N}!} \sim \left(\frac{e\mathrm{V}}{\mathrm{N}}\right)^{\mathrm{N}} = \left(\frac{e}{\rho}\right)^{\mathrm{N}},$$

where ρ is the density.

Let us calculate the logarithm of the partition function:

$$\log \mathrm{Z} = \mathrm{N}\left(\frac{3}{2}\log\frac{2m\pi}{\beta} - \log\rho + 1\right).$$

Now we can differentiate to find the total energy:

$$\mathrm{E} = -\frac{\mathrm{d}\log \mathrm{Z}}{\mathrm{d}\beta} = \frac{3}{2}\mathrm{NT}.$$

Pressure of an Ideal Gas and Fluctuations

Rewriting the expression

$$\mathrm{S} = \beta\mathrm{E} + \log \mathrm{Z},$$

we have

$$\mathrm{E} - \mathrm{TS} = -\mathrm{T}\log \mathrm{Z}.$$

Ignoring the Boltzmann constant, this is the **Helmholtz free energy**.

By definition of pressure (force divided by area) and the work-energy theorem,

$$\mathrm{dE} = -\mathrm{PA}\mathrm{d}x = -\mathrm{PdV}.$$

Then we can write the pressure as minus the derivative, with respect to the volume, of the Helmholtz free energy, keeping the temperature fixed:

$$\begin{aligned} P &= -\left.\frac{\partial E}{\partial V}\right|_S = -\left.\frac{\partial E}{\partial V}\right|_T + \left.\frac{\partial E}{\partial S}\right|_V \left.\frac{\partial S}{\partial V}\right|_T \\ &= -\left.\frac{\partial E}{\partial V}\right|_T + T\left.\frac{\partial S}{\partial V}\right|_T \\ &= -\left.\frac{\partial}{\partial V}(E - TS)\right|_T . \end{aligned}$$

Moreover,

$$P = T\left.\frac{\partial \log Z}{\partial V}\right|_T .$$

In statistical mechanics, the standard deviation is also called the **fluctuation**. Let us compute the fluctuation of the average energy:

$$\begin{aligned} \langle E^2\rangle - \langle E\rangle^2 &= \sum \frac{1}{Z} e^{-\beta E_i} E_i^2 - \langle E\rangle^2 \\ &= \frac{1}{Z}\frac{\partial^2 Z}{\partial \beta^2} - \frac{1}{Z^2}\left(\frac{\partial Z}{\partial \beta}\right)^2 \\ &= \frac{\partial}{\partial \beta}\left(\frac{1}{Z}\frac{\partial Z}{\partial \beta}\right) \\ &= -\frac{\partial \langle E\rangle}{\partial \beta} \\ &= \frac{\partial \langle E\rangle}{\partial T} T^2 \\ &= C_v T^2 . \end{aligned}$$

The derivative of the average energy E with respect to the temperature T is called **heat capacity**, denoted by C_v. It tells us how much do you have to change the energy to make a certain change in temperature.

Weakly Interacting Gases, Heat, and Work

The kinetic energy of a molecule, under the effect of a force from every other molecule, can be written as

$$E = \sum_n \frac{p^2}{2M} + \sum_{n>m} U(|x_n - x_m|),$$

where p is the momentum, x is the position, M is the mass, and U is the potential energy.

If we fix one particle and integrate over another one, this gives us

$$\int d^3x\, U(|x|) =: U_0.$$

Now, if we integrate the position of the starting molecule, this gives a factor of volume. Hence,

$$\int d^3x_1 d^3x_2\, U(|x_1 - x_2|) = VU_0.$$

Let us compute the partition function

$$\begin{aligned}
\mathrm{Z} &= \int \frac{\mathrm{d}p\mathrm{d}x}{\mathrm{N}!} e^{-\beta\frac{p^2}{2\mathrm{M}}} e^{-\beta \mathrm{U}(x)} \\
&= \int \frac{\mathrm{d}p}{\mathrm{N}!}\, e^{-\beta\frac{p^2}{2\mathrm{M}}} \int \mathrm{d}x\, e^{-\beta \mathrm{U}(x)} \\
&= \int \frac{\mathrm{d}p\mathrm{V}^{\mathrm{N}}}{\mathrm{N}!}\, e^{-\beta\frac{p^2}{2\mathrm{M}}} \int \mathrm{d}x\, \frac{e^{-\beta \mathrm{U}(x)}}{\mathrm{V}^{\mathrm{N}}} \\
&\approx \mathrm{Z}_0(\beta) \int \frac{\mathrm{d}x}{\mathrm{V}^{\mathrm{N}}} \left[1-\beta \mathrm{U}(x)\right] \\
&= \mathrm{Z}_0(\beta) \left(1-\beta \int \frac{\mathrm{d}x}{\mathrm{V}^{\mathrm{N}}} \sum_{n>m} \mathrm{U}(|x_n - x_m|)\right) \\
&= \mathrm{Z}_0(\beta) \left(1-\beta \frac{\mathrm{N}(\mathrm{N}-1)}{2} \int \frac{\mathrm{d}x_1 \mathrm{d}x_2}{\mathrm{V}^{\mathrm{N}}}\, \mathrm{U}(|x_1 - x_2|)\mathrm{V}^{\mathrm{N}-2}\right) \\
&\approx \mathrm{Z}_0(\beta) \left(1-\frac{\beta \mathrm{N}^2}{2\mathrm{V}} \mathrm{U}_0\right),
\end{aligned}$$

where $\mathrm{Z}_0(\beta)$ is the partition function for the ideal gas.

Taking the logarithm,

$$\log \mathrm{Z} = \log \mathrm{Z}_0 + \log\left[1-\frac{\beta \mathrm{N}^2}{2\mathrm{V}} \mathrm{U}_0\right] \approx \log \mathrm{Z}_0 - \frac{\beta \mathrm{N}^2}{2\mathrm{V}} \mathrm{U}_0.$$

Calculating the energy,

$$\mathrm{E} = -\frac{\partial \log \mathrm{Z}}{\partial \beta} = \frac{3}{2}\mathrm{NT} + \frac{\mathrm{N}^2}{2\mathrm{V}} \mathrm{U}_0 = \frac{3}{2}\mathrm{NT} + \frac{\rho}{2}\mathrm{NU}_0.$$

And calculating the pressure,

$$\mathrm{P} = \mathrm{T}\, \frac{\partial \log \mathrm{Z}}{\partial \mathrm{V}}\bigg|_{\mathrm{T}} = \frac{\mathrm{NT}}{\mathrm{V}} + \frac{\mathrm{N}^2}{2\mathrm{V}^2} \mathrm{U}_0 = \rho \mathrm{T} + \frac{1}{2}\rho^2 \mathrm{U}_0,$$

where we used the pressure of the ideal gas $\mathrm{PV} = \mathrm{NT}$.

In this case, we are studying the range where the potential energy is much smaller than the kinetic energy.

By the first law of thermodynamics,

$$\mathrm{dE} = -\mathrm{PdV} + \mathrm{TdS} = -\mathrm{dW} + \mathrm{dQ},$$

where W is work and Q is heat. Is the heat an exact differential, i.e., is

$$\mathrm{dQ} = \mathrm{dE} + \mathrm{PdV}?$$

Let us assume that

$$\frac{\partial \mathrm{Q}}{\partial \mathrm{E}} = 1, \quad \frac{\partial \mathrm{Q}}{\partial \mathrm{V}} = \mathrm{P}.$$

We use the curl test. Since

$$\frac{\partial^2 \mathrm{Q}}{\partial \mathrm{V} \partial \mathrm{E}} = 0, \quad \frac{\partial^2 \mathrm{Q}}{\partial \mathrm{E} \partial \mathrm{V}} = \frac{\partial \mathrm{P}}{\partial \mathrm{E}} \neq 0,$$

the heat is *not* an exact differential. It depends on how you move from one place to another. In other words, the amount of heat that you have to put into a system in order to change from one state to another state depends on the path that you take.

Entropy Vs. Reversibility

Consider the harmonic oscillator. The energy is

$$\frac{p^2}{2m} + \frac{kx^2}{2},$$

where k is the spring constant.

The partition function is

$$\mathrm{Z} = \int e^{-\beta \frac{p^2}{2m}} \, \mathrm{d}p \int e^{-\beta \frac{kx^2}{2}} \, \mathrm{d}x.$$

Changing the variables $q^2 = \frac{p^2\beta}{2m}$ and $y^2 = \frac{x^2\beta k}{2}$,

$$\mathrm{Z} = \sqrt{\frac{2m\pi}{\beta}} \sqrt{\frac{2\pi}{\beta k}} = 2\pi \sqrt{\frac{m}{k}} \frac{1}{\beta} = \frac{2\pi}{\omega} \frac{1}{\beta},$$

with $\omega = \sqrt{\frac{k}{m}}$ is the frequency.

Hence,

$$\log \mathrm{Z} = c - \log \beta,$$

and

$$\mathrm{E} = -\frac{\partial \log \mathrm{Z}}{\partial \beta} = \frac{1}{\beta} = \mathrm{T}.$$

For the quantum oscillator, the energy is quantised, which means that it comes in discrete multiples of Planck's constant times the frequency. So the energy of an oscillator is $n\hbar\omega$ and we can rewrite the partition function as

$$\mathrm{Z} = \sum_n e^{-\beta n \hbar \omega} = \sum_n (e^{-\beta \hbar \omega})^n = \frac{1}{1 - e^{-\beta \hbar \omega}}.$$

The energy is

$$\mathrm{E} = -\frac{1}{\mathrm{Z}} \frac{\partial \mathrm{Z}}{\partial \beta} = \frac{\hbar \omega e^{-\beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}}.$$

Let us see what happens when the temperature is high. Since this is equivalent to low values of β, we can approximate the energy by

$$\mathrm{E} \approx \frac{\hbar \omega}{\beta \hbar \omega} = \frac{1}{\beta} = \mathrm{T}.$$

In plain words, quantum systems become classical when the temperature is very high. That's because the energy is high and the quantisation of energy becomes unimportant.

At low temperatures,

$$\mathrm{E} \approx \hbar \omega e^{-\beta \hbar \omega}.$$

That means that quantum mechanics tends to suppress the energy of an oscillator when the temperature is low enough.

Where is the crossover, the critical point that goes from quantum to classical? The crossover happens when $\beta\hbar\omega = 1$. Indeed,

$$\begin{cases} \beta \hbar \omega > 1 & \text{quantum,} \\ \beta \hbar \omega < 1 & \text{classical.} \end{cases}$$

Rewriting, the crossover happens when $\hbar\omega = \mathrm{T}$. Notice that the stiffness of the oscillator appears in the frequency ω.

Chaotic systems are those where two trajectories that start close to each other, eventually depart and spread out. This is why the second law of thermodynamics does not contradict Liouville's theorem. If your system starts with a probability distribution that looks like a 'blob', it will eventually fill out the whole phase space 'like fairy floss'. The volume stays the same.

Reversibility means that a phase point will eventually go back to the same point. That is, you can return to the original configuration.

Entropy, Reversibility, and Magnetism

The **Poincaré recurrence theorem** states that after sufficient long (but finite) time, a system will almost certainly return to a state arbitrarily close to their initial state.

Magnets resemble a lot of other things in nature and they are mathematically simple enough to study and interesting enough to exhibit features like phase transition.

The simplest magnet is one that either points up or down. Think, e.g., atoms in a crystal.

Let μ be the **magnetic moment** (related to the strength of the magnet) and H be the **magnetic field**.

If the first spin is up, we say that $\sigma(1) = +1$. If the second spin is down, we set $\gamma(2) = -1$. We set n to be the number of atoms pointing up and m the number of magnets pointing down. Then we may write the energy as

$$\mathrm{E} = (n-m)\mu\mathrm{H}.$$

The number of configurations (states) with this value of energy is

$$\frac{\mathrm{N}!}{n!m!}.$$

Note that $\mathrm{N} = n + m$.

The partition function is

$$\begin{aligned}
\mathrm{Z} &= \sum_{\substack{n,m \\ n+m=\mathrm{N}}} e^{-\beta\mu\mathrm{H}(n-m)} \\
&= \sum_{n} \frac{\mathrm{N}!}{n!(\mathrm{N}-n)!} x^n y^m \\
&= (x+y)^{\mathrm{N}} \\
&= \left(\frac{e^{-\beta\mu\mathrm{H}} + e^{\beta\mu\mathrm{H}}}{2}\right)^{\mathrm{N}} 2^{\mathrm{N}} \\
&= 2^{\mathrm{N}}(\cosh(\beta\mu\mathrm{H}))^{\mathrm{N}},
\end{aligned}$$

with $x := e^{-\beta\mu\mathrm{H}}$ and $y := e^{\beta\mu\mathrm{H}}$.

Magnetisation is defined as

$$\mathrm{M} = \left\langle \frac{n-m}{\mathrm{N}} \right\rangle.$$

It is the bias for each particle whether it's up or down.

Using the magnetisation, we can write the (average) energy

$$\mathrm{E} = \mathrm{NM}\mu\mathrm{H},$$

which can be computed

$$\mathrm{E} = -\frac{\partial \log \mathrm{Z}}{\partial \beta} = -\mathrm{N}\frac{\sinh(\beta\mu\mathrm{H})}{\cosh(\beta\mu\mathrm{H})}\mu\mathrm{H}.$$

The magnetisation is

$$\mathrm{M} = -\tanh(\beta\mu\mathrm{H}).$$

Notice that this model does not have a phase transition. The simplest one with phase transition is the Ising model in two dimensions.

Suppose that the energy is given by

$$\mathrm{E} = -j\sigma(1)\sigma(2).$$

Here the system is composed of just two points on the lattice. The minus sign denotes a system that favours being parallel.

If each particle is interacting with its neighbour,

$$\mathrm{E} = -j\sum \sigma(i)\sigma(i+1).$$

Notice that if we change every $\sigma(i)$ to $-\sigma(i)$, this does *not* change the energy. This is what it means to have a symmetry. Notice that in this case there is no bias to be up or down.

The Ising Model

Focusing on just one spin, the partition function is given by

$$\mathrm{Z} = \sum e^{\beta j \sigma} = e^{\beta j} + e^{-\beta j} = 2\cosh(\beta j).$$

The energy is

$$\mathrm{E} = -\frac{1}{\mathrm{Z}}\frac{\partial \mathrm{Z}}{\partial \beta} = -j\tanh(\beta j).$$

The average spin (or magnetisation) is simply

$$\mathrm{M} = \langle \sigma \rangle = -\tanh(\beta j).$$

In the (one-dimensional) Ising model, instead of thinking about the spins themselves, we consider the link between the spins. If we know that the first spin is up, then knowing $\mu_1 = \sigma_1\sigma_2$ is enough to determine the second spin. Then knowing $\mu_2 = \sigma_2\sigma_3$ is enough to determine σ_3, since $\sigma_2^2 = 1$. And so on.

This means that the energy is made out of these 'bond variables', meaning the relationship between neighbours. So we can write the energy as

$$\mathrm{E} = -j\sum_i \mu_i.$$

These bonds are all independent. Their values are either $+1$ if they are aligneded or -1 if they are anti-aligneded.

The partition function is

$$\mathrm{Z} = 2\sum_{\mu} e^{-j\sum_i \beta\mu_i} = 2(\cosh(\beta j))^{\mathrm{N}-1},$$

which is the same as the partition function for a simple magnet.

The magnetisation is again

$$\langle \mu \rangle = -\tanh(\beta j).$$

What happens if the ith spin is up? Is this information going to affect a spin further in the line, say the $i+n$ element? Quantifying this question,

$$\begin{aligned}\langle \sigma_i\sigma_{i+n}\rangle &= \langle \sigma_i\sigma_{i+1}\sigma_{i+1}\dots\sigma_{i+n-1}\sigma_{i+n-1}\sigma_{i+n}\rangle \\ &= \langle \mu_1\mu_2\dots\mu_{n-1}\rangle \\ &= (-\tanh(\beta j))^{n-1}.\end{aligned}$$

This means that correlation goes down as you go further in the line.

High temperature (low β) means low 'fidelity', you can't hear your neighbour, you get no information from them.

Now, let us move to the d-dimensional lattice. We use mean-field approximation. Every site is surrounded by $2d$ neighbours. The neighbours define a field whose fluctuation is much smaller than its average value.

Suppose that there is a bias in the system and denote the average spin by $\bar{\sigma}$. To compute the energy, we approximate the sum of the neighbours by $2d\bar{\sigma}$:

$$\mathrm{E} = -j\sigma \sum_{\text{neighbours}} \sigma = -2dj\sigma\bar{\sigma}.$$

This is called the **mean-field approximation**.

Using the computations above, the average of the initial spin is

$$\langle \sigma \rangle = -\tanh(2dj\beta\bar{\sigma}).$$

By translation-invariance and since every point is roughly the same as every other one, a self-consistent field theory is to say that $\langle \sigma \rangle = \bar{\sigma}$. To solve this equation, set $y = 2dj\beta\bar{\sigma}$. By plotting, one could see that at high temperatures, $y = 0$. At low temperatures the system has an overall global average of spin pointing all in the same direction. This is magnetisation. The critical temperature is $\mathrm{T} = 2dj$.

```r
# Monte Carlo simulation of a 2D Ising model using the Metropolis algorithm

library(ggplot2)

# Parameters
size = 10 # Width of square lattice
T = 2.5   # Temperature in units of epsilon/k

# Initialise to a random array
initialise <- function(size) {
  matrix(sample(c(-1,1), size*size, replace = TRUE),
              nrow = size, ncol = size)
}

# Colour a square according to s values
coloursquare <- function(s) {
  df <- expand.grid(i = 1:nrow(s), j = 1:ncol(s))
  df$spin <- as.vector(s)
  p <- ggplot(df, aes(x = j, y = i, fill = factor(spin))) +
    geom_raster() +
    scale_fill_manual(values = c("-1" = "black", "1" = "white")) +
    coord_fixed() +
    theme_void() +
    theme(legend.position = "none", plot.title = element_text(hjust = 0.5)) +
    labs(
      title = paste("2D Ising Model (Size = ", size, ", T = ", T, ")", sep = "")
    )
  print(p)
}

# Compute DeltaU of flipping a dipole (note periodic boundary condition)
DeltaU <- function(s,i,j) {
  top    <- if (i == 1)    s[size, j]  else s[i-1, j]
  bottom <- if (i == size) s[1,j]      else s[i+1, j]
  left   <- if (j == 1)    s[i, size]  else s[i, j-1]
  right  <- if (j == size) s[i, 1]     else s[i, j+1]
  2 * s[i,j] * (top + bottom + left +right)
}

s <- initialise(size)

for(iteration in 1:(100 * size^2)) {
  i <- sample(1:size, 1)          # Choose a random row number
  j <- sample(1:size, 1)          # and a random column number
  Ediff <- DeltaU(s,i,j)          # Compute DeltaU of hypothetical flip

  if (Ediff <= 0) {               # If flipping reduces the energy
    s[i,j] = -s[i,j]              # then flip it
  }
  else if (runif(1) < exp(-Ediff/T)) {  # otherwise the Boltzmann factor
      s[i,j] = -s[i,j]                  # gives the probability of flipping
  }

  if (iteration %% (size^2) == 0) {
    coloursquare(s)
    Sys.sleep(0.1)
```

```
  }
}
```

Liquid-Gas Phase Transition

Let us recall the Ising model, but adding an external field.

Each lattice site i holds a spin variable $\sigma_i \in \{+1, -1\}$, representing an "up" or "down" spin state.

The energy function is given by

$$\mathrm{E} = -\mathrm{J} \sum_{\langle i,j \rangle} \sigma_i \sigma_j - \mathrm{H} \sum_i \sigma_i.$$

Notice that parallel spins (i.e., $\sigma_i \sigma_j = 1$) form an "unbroken bond". Flipping a spin to create an anti-parallel pair costs 2J in energy.

Here, H is the external field, which acts on individual sites, biasing spins to aligned with the external magnetic field.

Rather than computing interactions across all individual neighbour configurations, the mean field approximation replaces explicit neighbour-neighbour interactions with an effective average field produced by all $2d$ neighbours. The self-consistency equation is given by

$$\bar{\sigma} = \tanh\left(\beta\left(2d\mathrm{J}\bar{\sigma} + \mathrm{H}\right)\right).$$

Setting $\mathrm{H} = 0$ reduces the equation to analysing the intersection of the linear function $y \cdot \frac{\mathrm{T}}{2d\mathrm{J}}$ and $\tanh(y)$.

At high temperatures, the only solution is $\bar{\sigma} = 0$, i.e., no magnetisation.

The critical temperature occurs when the linear slope at the original equals one, yielding the mean-field critical temperature $\mathrm{T}_c = 2d\mathrm{J}$.

At low temperatures, the linear function intersects $\tanh(y)$ at non-zero values, yielding two symmetric solutions. This represents **spontaneous magnetisation**, where an infinitesimal magnetic field breaks symmetry to select an overall "up" or "down" state.

Let us consider the phase transitions in the (T, H) plane. Below the critical temperature, crossing $\mathrm{H} = 0$ causes a jump in average magnetisation between net-negative and net-positive states. This is a signature of phase transition.

Above the critical temperature, varying H smoothly changes the magnetisation without jumps or distinct phase transitions.

A system can transform continuously from a negative to positive phase without experiencing a jump by routing around the critical point $(\mathrm{H} = 0, \mathrm{T} = \mathrm{T}_c)$ by varying, for example, the pressure.

References

- Leonard Susskind's Theoretical Minimum.
- Daniel V. Schroeder. An Introduction to Thermal Physics.
- L. D. Landau & E. M. Lifshitz. Statistical Physics.